Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[14MA41]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

MATHEMATICS-IV

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions. Choosing at least two from each part.

PART - A

1. a. Find the root of the equation $\cos x = xe^x$ using regula-falsi method. Carryout four iterations. **05**

b. Use Newton-Raphson iterative method, find the real root of $x\log_{10}x=1.2$ corrent to four decimal places. **05**

c. Find the missing values in the following table : **04**

x:	45	50	55	60	65
y:	3.0	-	2.0	-	-2.4

2. a. Apply Newton's forward difference interpolation formula to find the cubic polynomial which takes the following values **05**

X	0	1	2	3
f(x)	1	2	1	10

b. Find the polynomial f(x) by using Lagrange's interpolation formula for **05**

x	0	1	2	5
f(x)	2	3	12	147

c. Use Newton's divided difference interpolation formula to find f(6) from **04**

x	3	7	9	10
f(x)	168	120	72	63

3. a. Use the Trapezoidal rule to estimate the integral $\int_0^2 e^{x^2}dx$ taking 10 intervals. **05**

b. Evaluate $\int_0^{\pi/2} e^{\sin\theta}d\theta$ correct to four decimal places by Simpson's 3/8th rule. Divide interval into 3 equal parts. **05**

c. A rocket is launched from the ground. Its acceleration is registered during the first 80 second and is given in the table below. Using Simpson's 1/3rd rule, find the velocity of the rocket at =80 seconds **04**

t(sec)	0	10	20	30	40	50	60	70	80
f(cm/sec^2)	30	31.63	33.34	35.47	37.75	40.33	43.25	46.69	50.67

4. a. Find by Taylors' series method the value of y at x=0.1 to five places of decimals from $\frac{dy}{dx}=x^2y-1,\ y(0)=1$. **05**

b. Use Rung-Kutta method of fourth order, solve $\frac{dy}{dx}=\frac{y^2-x^2}{y^2+x^2}$ with y(0) =1 at x =0.2. **05**

c. Apply Milnes' predictor corrector method to compute y(1.4) correct to four decimal places, given $\frac{dy}{dx}=x^2+\frac{y}{2}$ and the data y(1)=2, y(1.1)=2.2156, y(1.2)=2.4649, y(1.3)=2.7514 **04**

PART – B

5. a. Derive the necessary conditions for the existence of the derivative of f(z) in Cartesian from. (Cauch-Riemann equations) **05**

b. Show that z^n is analytic. Hence find its derivative. **05**

c. Find the conjugate harmonic of $v(r,\theta)=r^2\cos 2\theta - r\cos\theta + 2$ **04**

6. a. Discuss the conformal transformation $w=z^2$ **05**

b. Discuss the conformal transformation $w=e^z$ **05**

c. Find the bilinear transformation which maps the points z =1, i,-1 onto the points w=i,0,-1. **04**

7. a. State and prove Cauchy's integral formula for analytic function f(z). **05**

b. Find the Laurents' expansion of $f(z)=\frac{7z-2}{(z+1)z(z-2)}$ in the region 1<z+1<3 **05**

c. Determine the pole and the residue's at each pole of f(z)= $\frac{2z+4}{(z+1)(z^2+1)}$ **04**

8. a. A particle moves along the curve $x=t^3+1$, $y=t^2$, z =2t +3 where t is the time. Find the components of its velocity and acceleration at t=1 in the direction of i +j + 3k **05**

b. Find the directional derivative of $\phi=x^2yz+4xz^2$ at he point (1,2,-1) in the direction of the vector 2i-j-2k. **05**

c. Prove that div $\left(r^n\vec{r}\right)=(n+3)r^n$ Hence show that $\frac{\vec{r}}{r}$ is soleinoidal. **04**